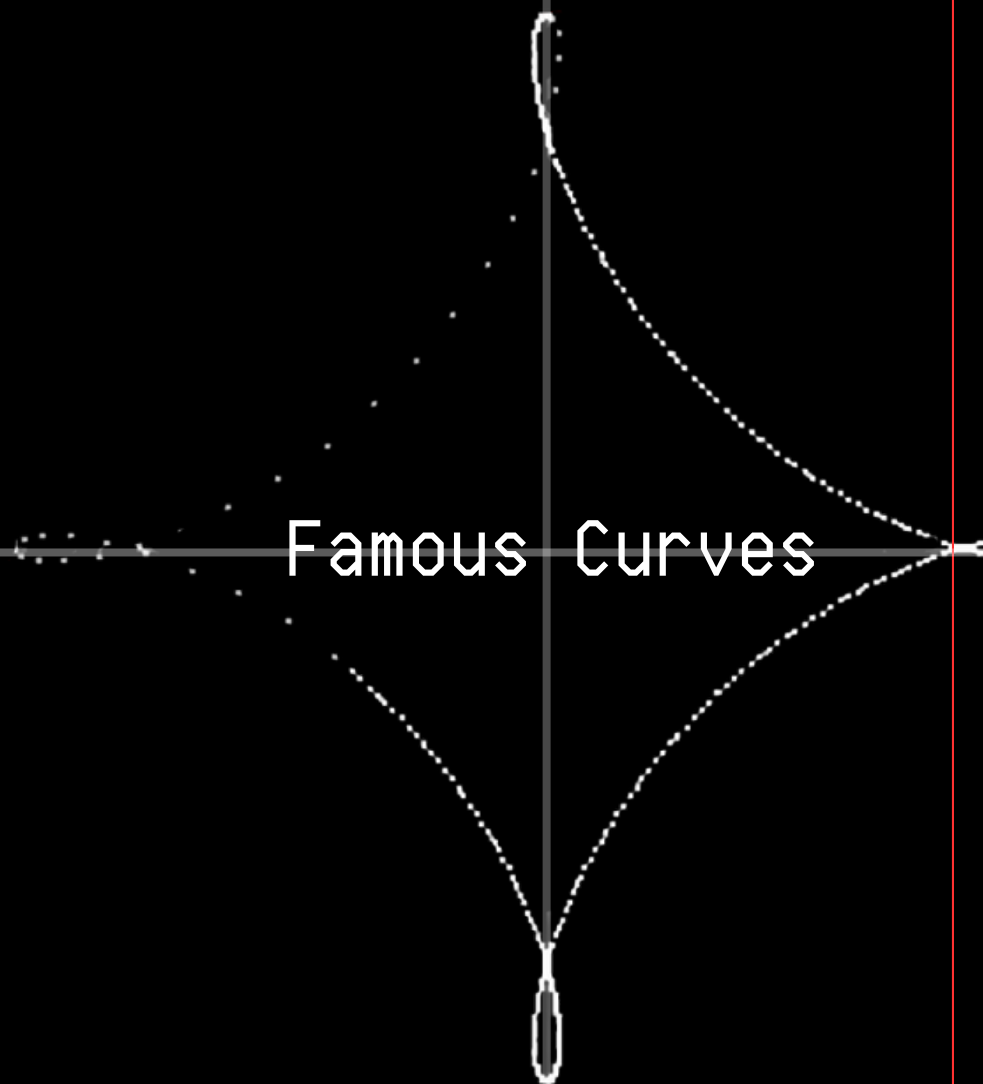


JSFX Plugin Suite 2026



User Manual

1. Introduction

This manual was developed as part of the project "Design and Development of Software for the Generation of Famous Curves for Electroacoustic Music Production and Education." The project was conducted at the School of Music of the Faculty of Humanities and Arts, National University of Rosario, as part of the 2024 Scientific Vocations Promotion Scholarship Program sponsored by the Argentine National Interuniversity Council (CIN).

1.1 About the Famous Curves Suite

The Famous Curves plugin suite was designed for musical composition and the teaching of electroacoustic music, with the purpose of generating spatial trajectories based on the so-called famous curves. The plugins were developed in the JSFX programming language to allow seamless integration with the REAPER digital audio workstation.

1.2 About the project

Each plugin generates, in real time and according to user-adjustable parameters, the coordinates of points belonging to a particular mathematical curve. The generated X and Y coordinates are normalized to values between -1 and 1. These coordinates can be used to control the movement of a sound source within a spatializer, or to modulate other musical parameters according to the source trajectory. The trajectory may be reproduced in real time or recorded as automation.

2. Basic Operation

2.1 Workflow

Each plugin consists of a curve generator and a curve display. The display allows the generated curve to be visualized in real time. The recommended workflow in REAPER is the following:

1. Insert a Famous Curves plugin together with a spatializer on the desired track or audio item.
2. Link parameters 11: X (Readout) and 12: Y (Readout) to the corresponding parameters of the spatializer. To do so:
 - Move the desired parameter on the spatializer plugin.
 - Open Param → Parameter Modulation/MIDI Link.
 - Verify that the selected parameter appears as Last Touched.
 - Enable Link from MIDI or FX Parameter.
 - Click (none) and select the desired Famous Curves plugin.
 - Choose parameter 11: X (Readout).
 - Use the Offset and Scale controls to ensure that the movement of the sound source inside the spatializer matches the motion shown in the curve display.
 - Repeat the same procedure for parameter 12: Y (Readout).

3. Adjust the curve parameters as desired, or load one of the available presets from REAPER's FX interface. Verify that the resulting trajectory matches the movement shown in the spatializer. If desired, automate any curve parameters over time (except the X and Y output parameters).
4. Record the spatializer automation.
5. Play back the recorded automation.

2.2 Interface

1. Preset selector.

2. Display

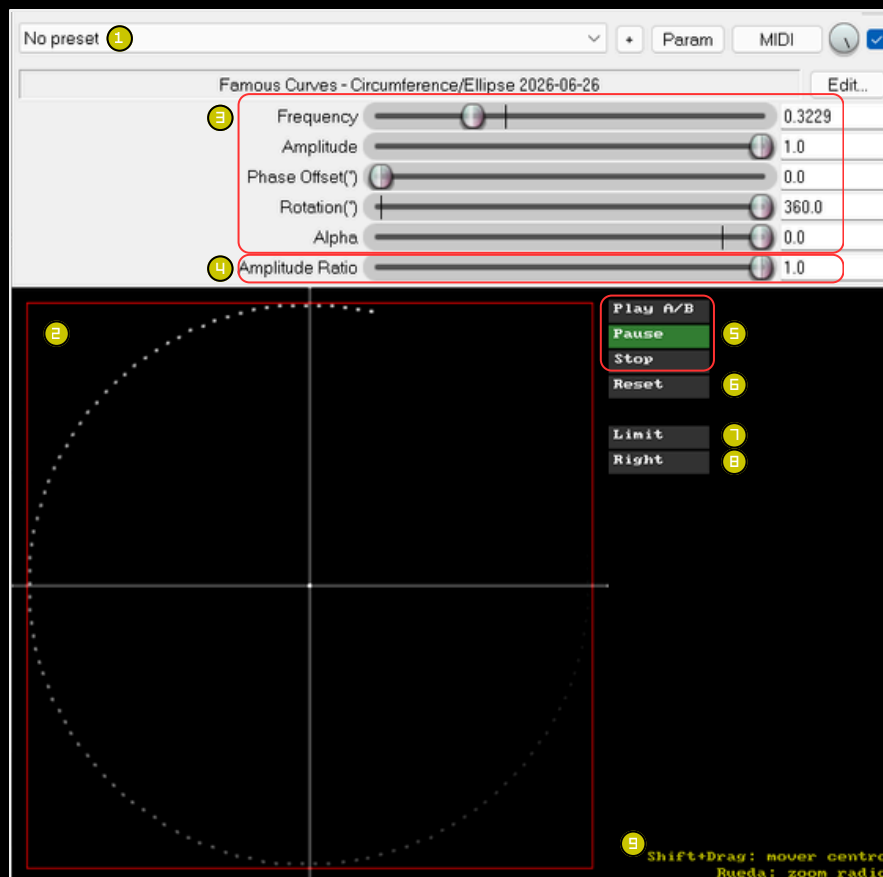
3. Parameters common to all plugins:

- **Frequency:** Controls the generation speed of the curve.
- **Amplitude:** Controls the overall size of the curve (the circle radius in this plugin).
- **Phase Offset (°):** Determines the starting phase of the curve generation.
- **Rotation (°):** Rotates the curve around its center.
- **Alpha:** Controls how long the curve trail remains visible.

4. Curve-specific parameters

5. Transport controls

6. Reset Button: Recenters the curve using the current parameter values.



7. Output Mode: Determines the behavior of the trajectory when it exceeds the boundaries of the display area.

- **Wrap mode:** The trajectory reappears on the opposite side, preserving continuous motion.
- **Limit mode:** Coordinates are clamped to the boundaries of the display area.

8. Direction button: changes the direction in which the curve is generated.

9. Comandos útiles con el mouse:

- Shift + Drag: Moves the center of the curve by adjusting Offset X and Offset Y.
- Mouse Scroll: Adjusts the curve amplitude.

2.3 Transport Controls

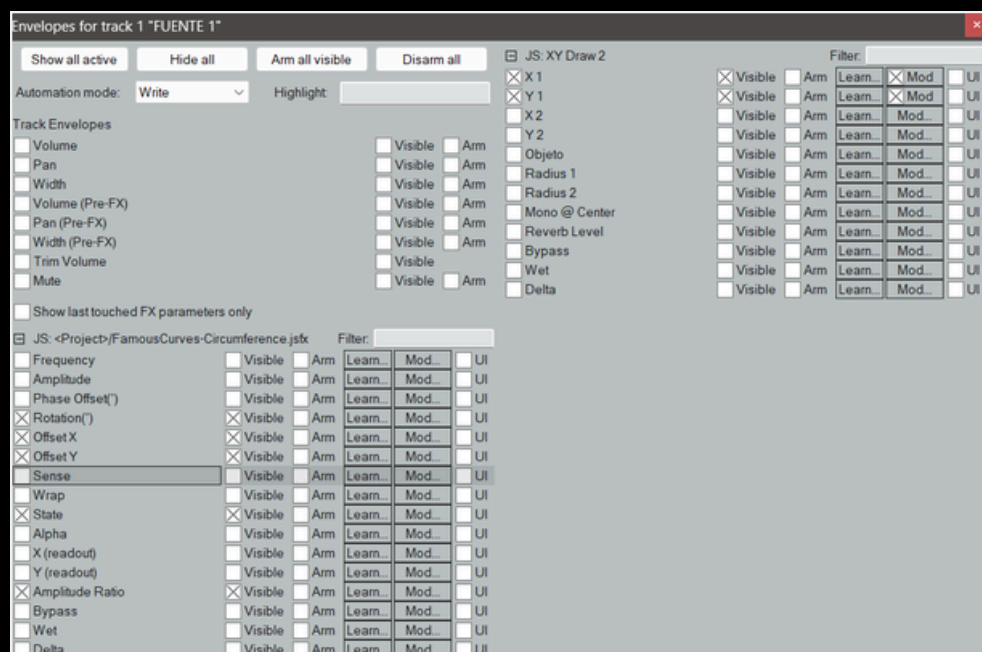
The transport provides four playback states: Play A, Play B, Pause, Stop. These states can be automated through parameter 7: State.

- **Play A/Play B:** Both modes reset the curve to the current Phase Offset and start playback. Switching between Play A and Play B makes it possible to retrigger the curve.
- **Pause:** Pauses the curve at its current position. Playback can be resumed from that point using either Play button.
- **Stop:** Resets the curve and stops its generation. This mode is useful for defining the initial position before playback.

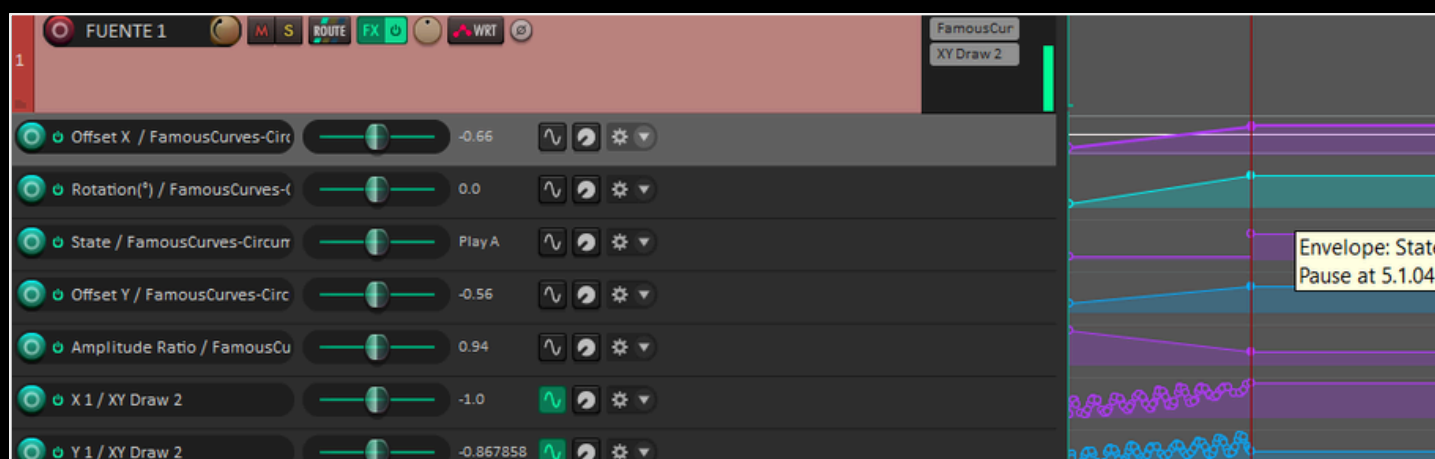
2.3 Recording

automation

Click the Track Envelopes button to open the list of automatable parameters for every plugin inserted on the track. Enable the checkbox next to each parameter to display its automation envelope.



Note that the vertical order of the envelopes corresponds to the order in which they were enabled. Although REAPER does not allow envelope reordering, community-developed plugins and extensions provide this functionality.



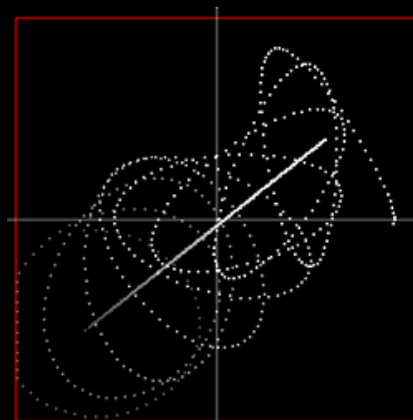
Once an envelope has been activated, its value is controlled by the automation lane rather than the plugin interface. Therefore, it is recommended to define the initial curve settings—such as frequency, amplitude, phase offset, rotation, and other parameters—before enabling automation.

To record automation from the spatializer:

- Set the track automation mode to Write.
- Make sure the desired spatializer parameters are armed for recording.
- Play the project.

Switching the automation mode to Touch allows the instantaneous parameter values to be edited directly from the plugin by placing the edit cursor at the desired time position. This workflow is particularly useful when positioning Offset X and Offset Y directly from the plugin using Shift + Drag.

As an example, if we were to reproduce the previously drawn automation we would obtain the trajectory drawn at the bottom right. The straight line represents the movement of the curve's center.



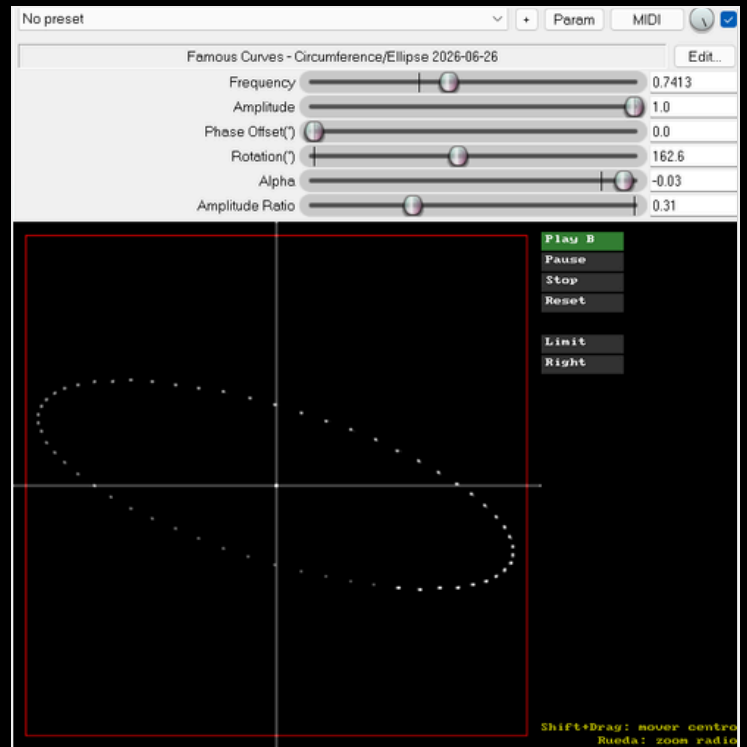
3. The Famous Curves Suite

3.1 Circle and Ellipse

The circle is generated by the uniform motion of a point around a fixed center while maintaining a constant distance from that center (the radius). The ellipse is generated similarly, but uses different amplitudes for the horizontal and vertical axes.

Dedicated parameters

-Amplitude Ratio: Defines the amplitude ratio between the X and Y axes. A value of 1 produces a perfect circle. Any other value produces an ellipse.



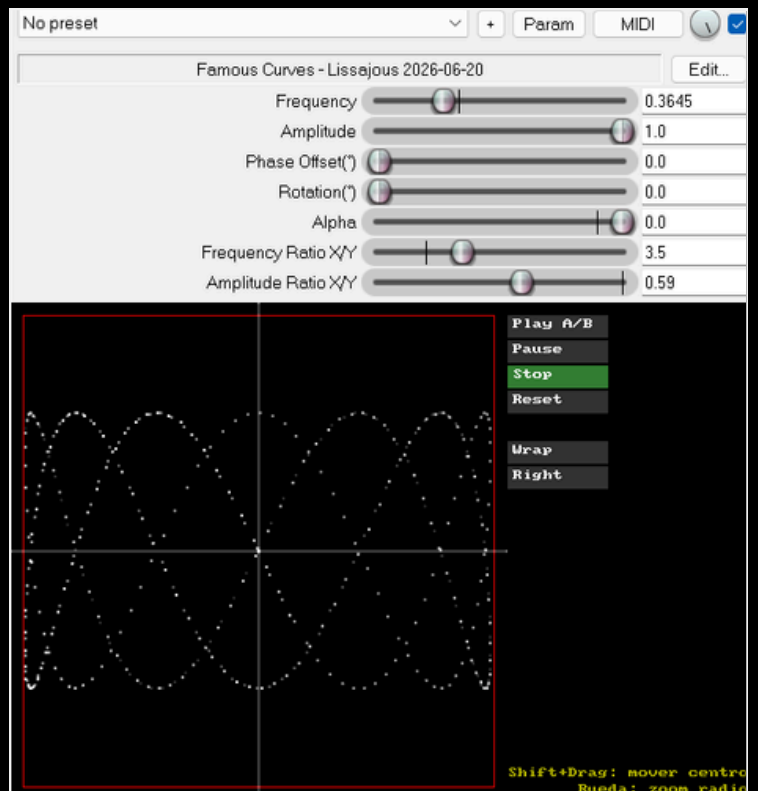
3.2 Lissajous

Lissajous curves are generated by combining two sinusoidal motions whose frequencies, amplitudes, and phase offsets can vary independently. The frequency ratio determines the number of lobes and the overall complexity of the resulting trajectory.

Dedicated parameters

-Frequency Ratio: Defines the ratio between the X and Y frequencies.
-Amplitude Ratio: Defines the ratio between the X and Y amplitudes.

In this plugin, **Phase Offset** shifts the oscillator generating the X values relative to the one generating the Y values, producing significant changes in the resulting curve. For example, A 1:1 frequency ratio with zero phase offset produces a **straight line** with pendulum-like motion. A 90° phase offset produces a circle. A 2:1 frequency ratio with zero phase offset produces a **lemniscate** (figure-eight), meanwhile a 45° phase offset produces an **exponential curve** with pendulum-like motion [see presets].



3.3 Spirograph

This plugin generates curves traced by a point attached to a circle of radius r rolling around a fixed circle of radius R . The point is located at a distance d from the center of the rolling circle. When the rolling circle moves outside the fixed circle, the result is an **epitrochoid**. When it rolls inside, the result is a **hypotrochoid**.

Dedicated parameters

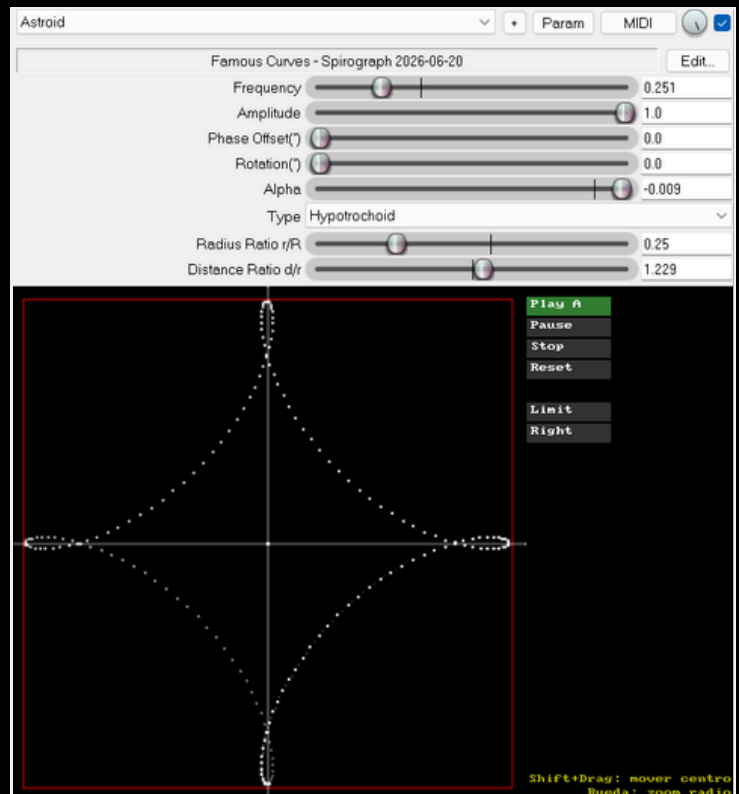
-Type: Selects either Hypotrochoid or Epitrochoid.

-Radius Ratio r/R : Defines the ratio between the radii of the two circles.

-Distance Ratio d/r : Defines the distance of the tracing point from the center of the rolling circle relative to its radius.

Special cases occur when $d/r = 1$, (meaning that the tracing point lies directly on the rolling circle): **Hypocycloids** and **epicycloids** respectively. Looking for other special cases we found again the circumference and the ellipse, but

also some new ones: **deltoid**, **astroid**, **cardioid**, **nephroid** and **Pascal's Limaçon** (see presets).



3.3 Rhodonea

This plugin generates the so-called **rhodonea** using a radial function in polar coordinates, where the distance from the origin varies periodically with the angle, producing a sequence of petals.

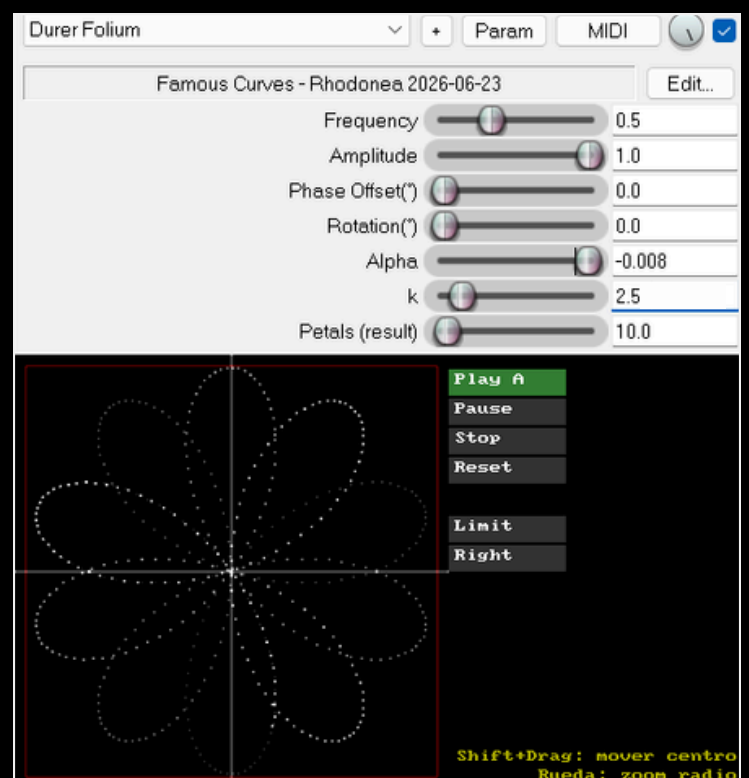
Dedicated parameters

- k : Determines the number of petals.

If k is an integer:

- Even: the curve has $2k$ petals.
- Odd: the curve has k petals.

If k is a rational number, determining the number of petals becomes more complex. For this reason, an additional parameter displays the resulting number of petals for the selected value of k .



3.5 Spirals

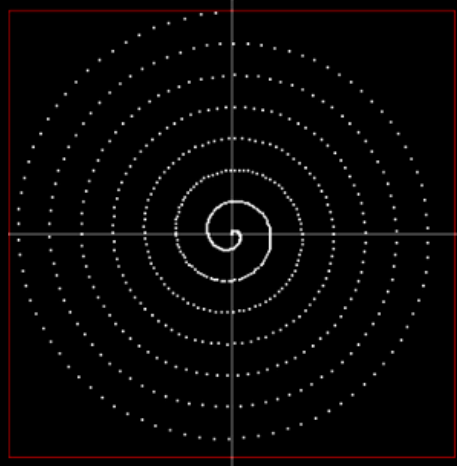
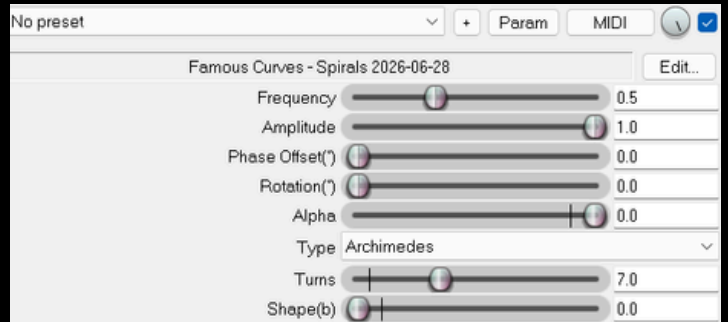
Spirals are plane curves whose radius varies continuously as the angle increases around a central point. Depending on the relationship between the radius and the angle, they may exhibit linear, exponential, hyperbolic, or other types of growth, resulting in trajectories with distinct geometric characteristics.

Dedicated Parameters

-Type: Selects the type of spiral to be generated.

-Turns: For the majority of spirals, defines the number of revolutions completed.

-Shape: Controls the degree to which the selected curve deviates from a circle. A value of 1 produces a circumference with a radius equal to **Amplitude**. Depending on the selected spiral, this parameter controls the radial growth or decay rate, radial oscillation, or the position of asymptotes.



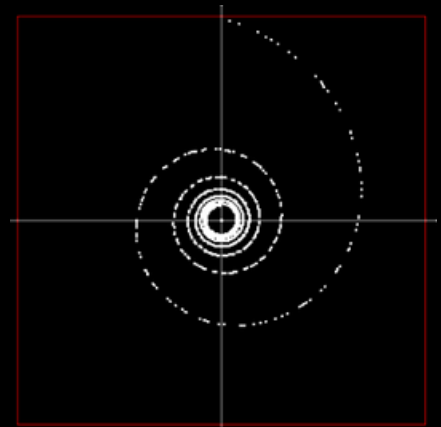
Shift+Drag: mover centro
Rueda: zoom radio

3.5.1 Archimedean Spiral

The radius increases linearly with the angle, producing a constant separation between consecutive turns. It is one of the simplest and most widely used spirals. **Shape** modifies the initial radius, reaching a radius of **Amplitude** in set **Turns**.

3.5.2 Hyperbolic Spiral

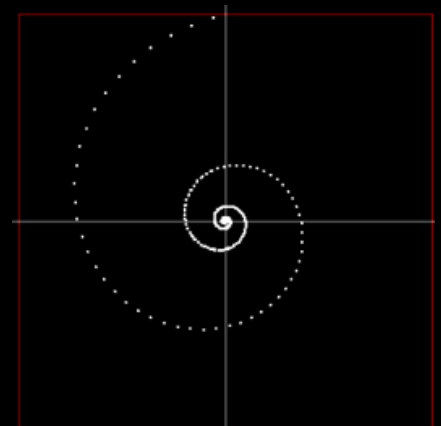
The radius decreases inversely proportional to the angle. The curve approaches the origin without reaching it, while its turns become progressively denser. It works best for **Shape** > 0.5, although it needs to be rescaled with **Amplitude**.



3.5.3 Logarithmic Spiral

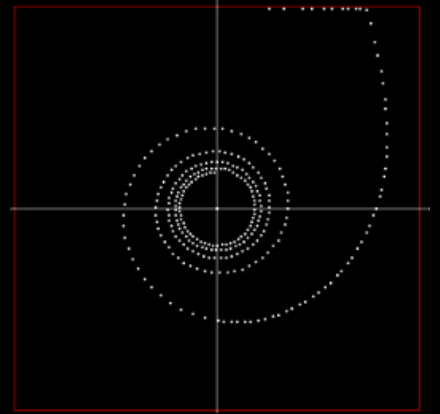
The radius grows exponentially with the angle, maintaining a constant angle between the tangent and the radius. It is common in natural forms such as seashells, galaxies, and hurricanes.

Being logarithmic, most of the path is covered in the final turns, so a greater number of turns only means it will stay closer to the center for longer. This is modifiable via **Shape**.



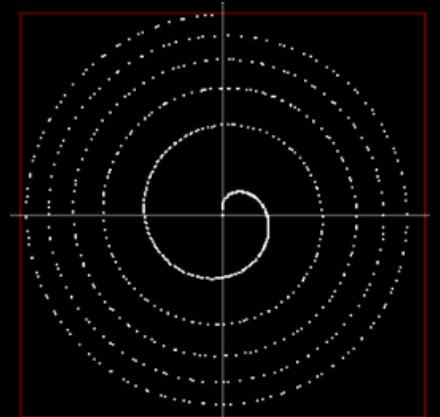
3.5.4 Lituus Spiral

The radius decreases with the square root of the angle. As the curve asymptotically approaches the origin, the turns become increasingly compact. **Shape** allows manipulating the distance to the center. It is usually necessary to **limit** it and rescale it with **Amplitude**.



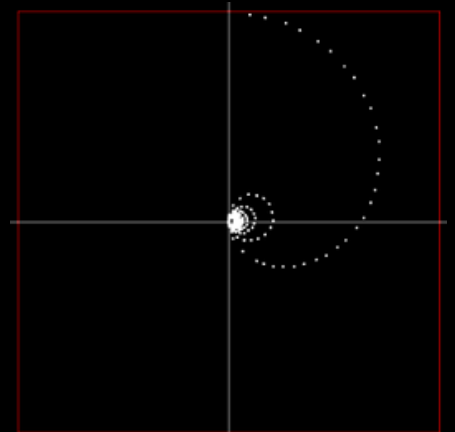
3.5.5 Fermat's Spiral

The radius increases with the square root of the angle. Its turns gradually separate as the spiral moves away from the center, generating smoother growth than that of the Archimedean spiral. **Shape** allows manipulating the distance to the center.



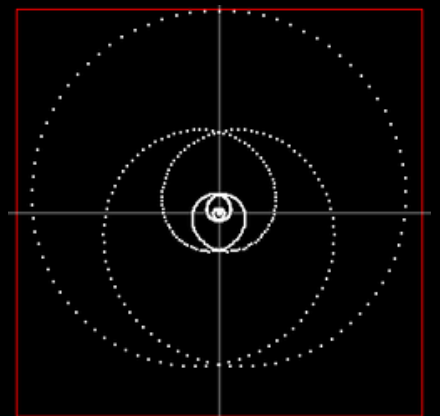
3.5.6 Cochleoid Spiral

The cochleoid combines a spiral behavior with loops around the origin. It has a branch that approaches the center, forming increasingly smaller loops around it. **Shape** produces a particular behavior, causing the loops that form to rotate around the center as they generate.



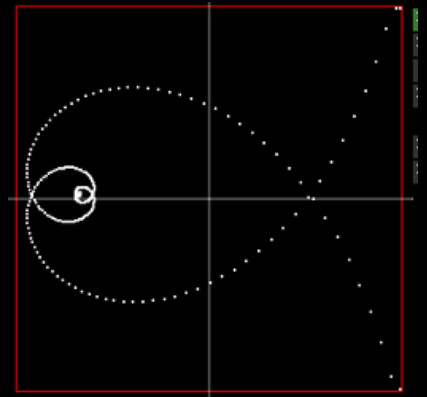
3.5.7 Poincaré Spiral (Hyperbolic Secant)

The radius decreases following the sech (hyperbolic secant) function, forming two branches that coil in opposite directions and converge asymptotically toward the origin. **Shape** controls how quickly it decays, opening up the turns and making the path more useful. It is advisable to increase the number of turns enough to achieve closure of the figure, otherwise much of the trajectory is lost infinitesimally at the origin.



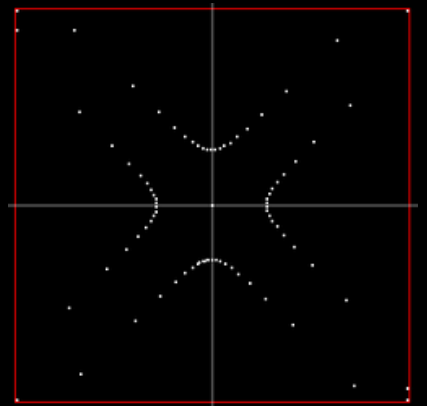
3.5.8 Poinot Spiral (Hyperbolic Cosecant)

The counterpart of the previous one, using the cosh (hyperbolic cosecant) function. Unlike the secant, the radius diverges at the center of the sweep instead of reaching a finite maximum, so the curve moves far away from the origin before collapsing back toward the center on both sides. **Shape** controls the opening of the turns, but since its center is at the origin of coordinates, due to its shape it easily goes out of range. For this reason it is advisable to keep **Shape** < 0.6 approximately, **limit** it, rescale it with **Amplitude**, and offset it from the origin.



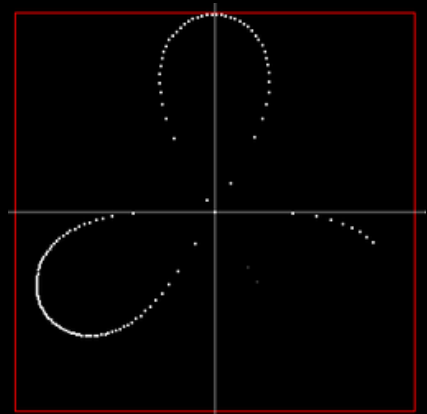
3.5.9 Epispiral

This is the result of the polar or circular inversion of the polar rose. It repeats the same pattern periodically. This curve is manipulated only with **Shape**, generating a greater number of branches. It tends toward infinity, so it must be limited with **Limit** and rescaled with **Amplitude**.



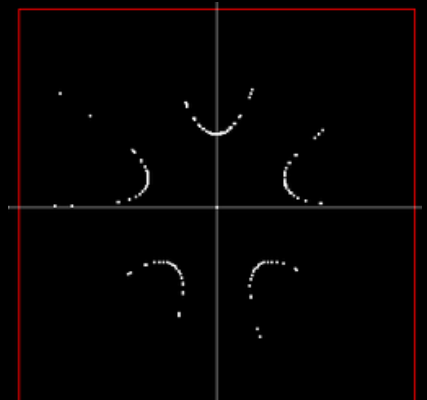
3.5.10 Sinusoidal Spiral (+)

Generates closed curves, shaped like lobes that repeat symmetrically around the origin. Depending on the value of **Turns**, it moves from a simple circle to a lemniscate or a multi-petaled rose. Experiment with **Shape**, which affects the shape of the petals in various ways. For **Shape** = 0.5, for example, the petals become thicker.



3.5.11 Sinusoidal Spiral (-)

The "open" counterpart of the previous one, generating branches that shoot off toward infinity instead of closed lobes, and therefore must be limited. Same considerations as above regarding **Shape** and **Turns**.



4. Presets

Each plugin includes a collection of presets intended to demonstrate the different curves available and provide useful starting points for creating new trajectories.

The preset names correspond either to the represented mathematical curve or to a particular configuration of the plugin parameters.

5. Source code

The complete source code of the Famous Curves suite is available under an open-source license and can be accessed through the project's official repository.